

Shafiq Ahmad

Lead Software Engineer

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Summary

Lead Software Engineer and **Software Architect** with **11+ years** designing and shipping **billion-scale distributed systems, event-driven microservices**, and data infrastructure across four continents. Redesigned a legacy pipeline from **18M → 1B messages/month** (50x throughput) using an **Actor Model** on Kafka and Kubernetes. Ingested **1 billion rows in 10 minutes** on a time-series rewrite, and cut vector-search latency by **60%**. End-to-end ownership from R&D PoC to production, spanning **on-prem, multi-cloud, IoT (1M MQTT msg/sec)**, and corporate analytics. **5+ years leading and mentoring teams** in **Agile / Scrum** environments; also **Developer Advocate & Pre-Sales Engineer** (GITECH Berlin).

Contributed to multi-region projects for a **Critical Infrastructure Energy** provider, ensuring scalability, reliability, and operational efficiency, with close collaboration between **Engineering, DevOps**, and **Business Stakeholders**. Also worked in pure **R&D**, reading **Distributed Consensus** and **SHA-256 Cryptography** in **Verilog** papers and implementing them.

Led engineering teams in both **startups** and **large organizations**, and equally comfortable designing **system architecture** as debugging performance issues at 3 AM or implementing the tricky parts myself. Motivated by learning and high impact, and thrive in fast-paced environments where **ownership** and **initiative** are encouraged.

Outside of tinkering with Software & LLM/AI, I enjoy hiking in the Alps and play some DOTA2 whenever I have time.

Experience

AlgoMinds GmbH *Düsseldorf, Germany* *Oct 2024 - June 2026*

Lead Software Engineer

Drove technical direction for Deployment Platform and a Siemens Energy Control System. Owned architecture decisions, trade-offs, and long-term system health. Filled in for our Team Manager in **Agile PI** planning, architecting team's roadmap for next sprints while balancing business priorities.

- Designed a data pipeline at **1M MQTT messages/sec** from **Bosch Rexroth IoT devices** through **AWS IoT Core** to an internal time-series database, including throughput tests and cloud cost estimation. Owned the data transport and storage layer end-to-end.
- Extended the Ignition L2 SCADA, with **Java/Kotlin** plugins for **Inductive Automation Ignition** and ported a legacy Java Time-Series DB client (custom stateful transport protocol) to .NET.
- Automated **CtrlX Bosch Rexroth** Snap application builds from Windows hosts via **Ubuntu WSL**, using it for local development, IoT deployment, certificate provisioning, and integration tests.
- Authored **Kubernetes** configuration for platform services and **Skaffold**-based remote development setup, enabling engineers fast iteration against real K8s environments.
- Shaped **DevSpells**, a Deployment Platform extending the **.NET Aspire** model with first-class **Microservices, Multi-Cloud**, and **On-Prem deployment** support, built on **Pulumi IaC** with database-agnostic persistence via .NET transaction scopes over **MongoDB**.
- Solved rootless **Docker-in-Docker** for **Azure DevOps** self-hosted agents, hardening against supply-chain attacks, by working through **SELinux, AppArmor**, device mounts, and WSL-vs-Debian kernel differences. Trade-off: accepted a small perf cost for stronger execution isolation to meet security compliance without breaking **CI/CD** reliability.
- Represented the company at **GITECH Berlin** as Developer Advocate and pre-sales engineer: demonstrated DevSpells live to prospective customers and partners, translating deep technical capabilities into business value in booth and 1:1 sessions.

Nyris GmbH Berlin, Germany Apr 2022 - Oct 2024

Senior Software Engineer

Shaped technical strategy across R&D and production, from **billion-scale data systems** to company-wide engineering standards. Accountable for long-term system health and the architectural trade-offs behind it. Mentored teammates via 1:1s, sprint planning, pair-programming, and collaborative code reviews; established standups, sprints, retrospectives, and an anonymous kudos culture for the team. Also shipped internal tooling and standards for code quality.

- Owned technical strategy for **Inter Cars onboarding**, the company's largest customer, navigating strict security and infrastructure constraints to define the **Kafka**-based data ingestion approach that became the template for subsequent enterprise deals.
- Architected the **Data Orchestrator (Kafka + KEDA + Rust workers + gRPC)** based Actor Model system, redesigning the pipeline from **18M to 1B messages/month**. Chose Kafka over RabbitMQ for durability guarantees, accepting higher operational complexity in exchange for 50x throughput gains.
- Led a reporting system with **TDengine** time-series DB and achieved **1B-row ingestion in 10 minutes**, replacing a legacy system that took hours for a fraction of the data.
- Implemented **Delta OR-Set CRDT** synchronization (based on "Efficient Synchronization of State-based CRDTs") for an in-house database. Reduced sync payload by approximately 90% via tombstone sets; drove leadership buy-in to adopt OR-Set as the foundation for all system data types.
- Co-developed a **SIMD/AVX** vector-stride kernel for embedding similarity search on .NET, generalizing fixed-length kernels to arbitrary-length inputs; benchmarked native allocators, Marshal, and unsafe pointer arithmetic across Intel CPU generations on GCP.
- Drove company-wide adoption of **Qdrant** over **MemSQL (SingleStore)** after a CTO-approved evaluation identified vector search as the main bottleneck for the 1B-scale mandate. Reduced vector search latency by approximately 60% and cut annual infrastructure cost 5x, from roughly EUR 60,000 to EUR 12,000 per year.

CELUS (Contunity GmbH) Munich, Germany Apr 2020 - Mar 2022

Senior Software Engineer

Defined the client-facing app architecture and built internal developer infrastructure and tooling in a fast-moving startup. Delivered revenue-generating platform for electronics design engineers.

- Defined the frontend technical direction: led the **Angular-to-React** migration (with **JointJS/Rappid** as SVG schematics canvas), established the **Redux** application architecture, and introduced a rigorous testing culture from **Jest** unit tests through **Playwright** E2E.
- Architected the **Express/TypeScript backend-for-frontend** gateway and contributed across **Python Flask Microservices**, **MongoDB**, **Airflow**, and **RabbitMQ**, shaping service boundaries for the platform.
- Built three org-wide internal developer tools: **Xandria** (search over internal docs via **Playwright + NestJS + Elasticsearch**), **Local Deployment** (one-command **Docker Swarm** dev environment), and a **Keycloak auth-proxy** pattern adopted as the company-wide standard.
- Created **Terraform + Ansible** automation for ephemeral **EC2** dev machines, improving onboarding speed and security posture. After detecting brute-force attacks on long-lived instances, compiled a security report that drove tightened access controls across the org.
- Helped grow the team: ran candidate interviews for senior frontend engineers and onboarded new hires onto the **React TypeScript** stack.

Oozou Pvt Ltd. Bangkok, Thailand Apr 2019 - Nov 2019

Software Engineer

Client-facing consultancy. Built and optimized **React/Next.js** apps for True Corporation (Thailand's largest ISP); forward-deployed for BANPU (South-East Asia energy) on **React Native**. Performed Lighthouse audits and introduced lazy image loading via Intersection API. Interviewed Full-Stack candidates.

Lead Software Engineer

- Led a 5-engineer team through a full-stack migration from PHP/Ember to **React, Node.js, TypeScript** introduced Jest unit and Puppeteer E2E testing.
- Delivered web and mobile for Ace Hardware and Epicor across iOS, Android, and UWP, including a **Xamarin + OpenCV face-detection timeclock app** (HAAR cascades via **C++** interop). Introduced Docker, **CircleCI CI/CD**, Detox/Puppeteer E2E, and GraphQL for versioned APIs.

eConceptions *Islamabad, Pakistan Jan 2015 - Dec 2015*

Associate Software Engineer

Early-career mobile and web engineering across **React Native, Ionic/Cordova, Angular**, and **ASP.NET/C#** for international clients including Careem (ride-hailing, Middle East) and Marhaba Malls (UAE eCommerce). Wrote native Cordova plugins to bring app animations close to 60fps.

AI-Augmented Engineering

Daily practitioner of AI-assisted engineering with **GitHub Copilot, Claude Code, OpenCode**, and **Pi Agent**, across **Claude Subscription** and **Local LLM Models**. Continuously evolving workflow built around custom skills, planning agents, and Ralph-loop-style plugins.

Skills

Languages: TypeScript/JavaScript, C#, Rust, Python, C++

Architecture: Software Architecture, Distributed Systems, Microservices, Event-Driven Architecture, Actor Model, REST, gRPC, GraphQL, CQRS, CRDTs, Distributed Computing, Workflow Orchestration, Data Modeling, Information Architecture, SaaS

Distributed Data: Kafka, RabbitMQ, AWS SQS, MQTT, TDengine, PostgreSQL, NoSQL, MongoDB, Redis, Elasticsearch, Qdrant, MemSQL/SingleStore, SIMD/AVX

Platforms: React.js, Next.js, Node.js, HTML, React Native, NestJS, Flask, .NET/ASP.NET Core, Entity Framework, Generative AI, Agentic AI, Anthropic, OpenAI, Computer Vision

Cloud & Infra: Cloud Native, Kubernetes, Docker, Terraform, Pulumi, Ansible, AWS, Azure, Azure DevOps, GCP, Cloud Security, CI/CD, GitHub Actions, GitLab CI, Secrets Management, Observability, OpenTelemetry, Grafana

Ways of Working: Developer Experience, Prototyping, Test Automation, Agile, Scrum, Sprint Planning

Soft Skills: Leadership, Mentoring, Teaching, Stakeholder Management, Cross-functional Collaboration, Public Speaking, Hiring & Interviewing, Conflict Resolution

Open Source

- [KurzSharp](#), .NET source-generator library (NuGet) for instant API scaffolding: generates REST, GraphQL, and gRPC endpoints plus a Blazor admin dashboard from annotated models.
- [Nyris.Crdt](#), C# CRDT library with distributed sync via gRPC source generators, implementing OR-sets, LWW registers, and partially replicated registries.
- [SHA256 Verilog](#), FPGA implementation of SHA-256 in Verilog.
- Upstream contributions to React Native (docs), yup (SameValue semantics for signed zero), and asm-dom-boilerplate (cross-platform hot reload).

Education

B.Sc. Electronics Engineering *Taxila, Pakistan Graduated 2014*

University of Engineering and Technology Taxila

Spoken Languages

Fluent: English, Urdu, Punjabi, Siraiki

Basic: German